

실습 1 Python 기초와 API 연결 환경 설정

Windows 10/11, PowerShell, Python 3.12.x

작업 폴더: `ai-python`, 가상환경: `ai-python/.venv`

1. Python 설치

1. Python 다운로드 페이지에서 Python 3.12.x 설치 파일을 받습니다.

Version	Operating system	Description	File size	Sigstore	SBOM	GPG	MD5 checksum
Gzipped source tarball	Source release		25.9 MB	.sigstore	SPDX	SIG	35c03f014408e26e2b06d576c19cac54
XZ compressed source tarball	Source release		19.6 MB	.sigstore	SPDX	SIG	0c0a6cce86d7611aed691d61e0599de0
macOS 64-bit universal2 installer	macOS	for macOS 10.13 and later	43.6 MB	.sigstore		SIG	8f4989592c9412e51fcad70273442c22
Windows installer (64-bit)	Windows	Recommended	25.7 MB	.sigstore	SPDX	SIG	5eddb0b6f12c852725de071ae681dde4
Windows installer (32-bit)	Windows		24.5 MB	.sigstore	SPDX	SIG	ff3e7e2582811ade44916d9c3d2a64b2
Windows installer (ARM64)	Windows	Experimental	25.0 MB	.sigstore	SPDX	SIG	37df52717faa6c514342b8d43cae7026
Windows embeddable package (64-bit)	Windows		10.6 MB	.sigstore	SPDX	SIG	fe8ef205f2e9c3ba44d0cf9954e1abd3
Windows embeddable package (32-bit)	Windows		9.5 MB	.sigstore	SPDX	SIG	93beb9042be23d7b5cd528fa15e8fa7
Windows embeddable package (ARM64)	Windows		9.9 MB	.sigstore	SPDX	SIG	60ce2b5cb5a4dfcd2873a323ffe541bc

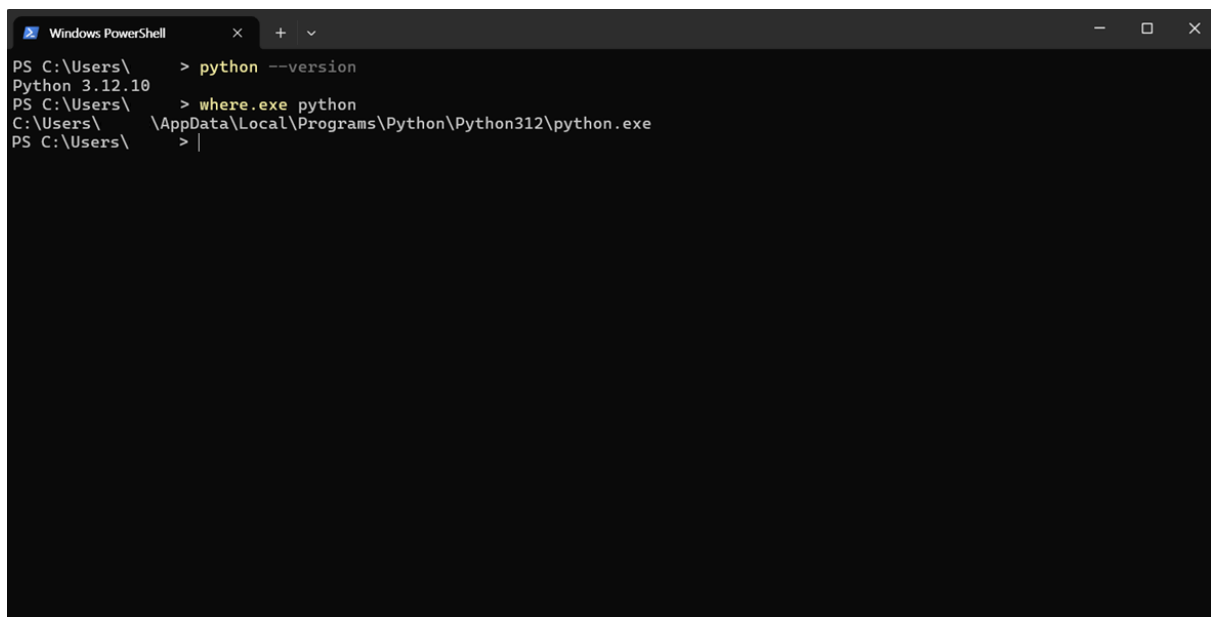
1. `Add python.exe to PATH` 를 선택합니다.
2. `Install Now` 를 선택합니다.



PowerShell을 새로 열고 실행합니다.

```
python --version  
where.exe python
```

예상 출력: Python 3.12.x, Python 설치 경로

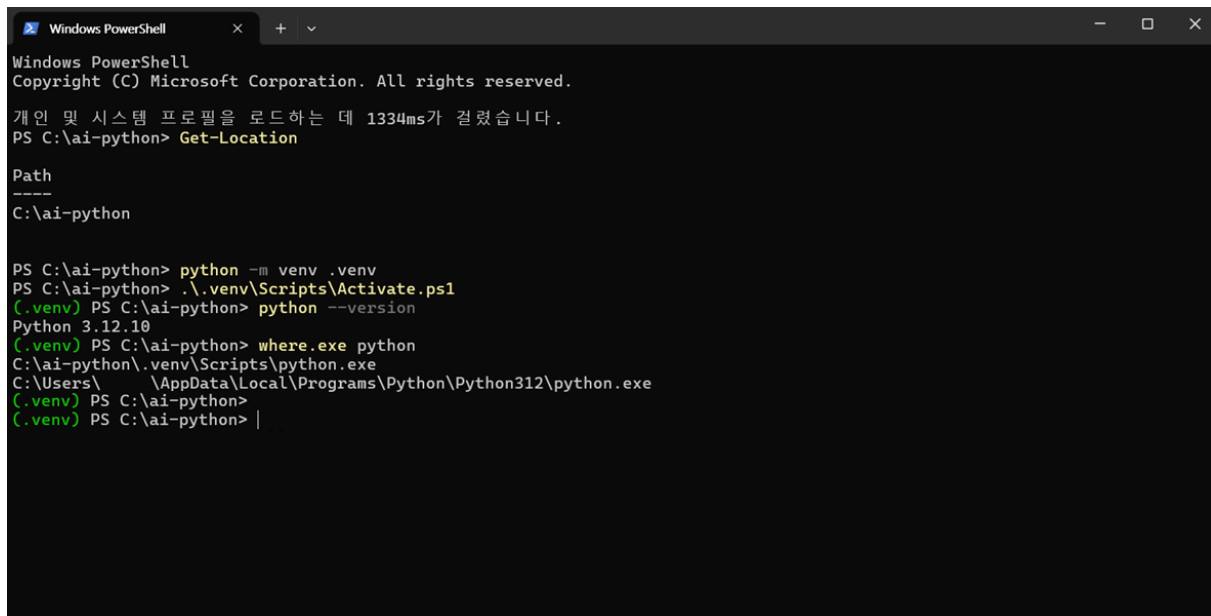


2. 가상환경

파일 탐색기에서 `ai-python` 폴더를 연 뒤 주소 표시줄에 `powershell` 을 입력합니다.

```
Get-Location
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python --version
where.exe python
```

확인 항목: `ai-python` 경로, `(.venv)` 표시, `.venv\Scripts\python.exe`



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

개인 및 시스템 프로필을 로드하는 데 1334ms가 걸렸습니다.
PS C:\ai-python> Get-Location

Path
----
C:\ai-python

PS C:\ai-python> python -m venv .venv
PS C:\ai-python> .\.venv\Scripts\Activate.ps1
(.venv) PS C:\ai-python> python --version
Python 3.12.10
(.venv) PS C:\ai-python> where.exe python
C:\ai-python\.venv\Scripts\python.exe
C:\Users\...\AppData\Local\Programs\Python\Python312\python.exe
(.venv) PS C:\ai-python>
(.venv) PS C:\ai-python> |
```

실행 정책 오류가 나오면 아래 명령을 실행합니다.

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\.venv\Scripts\Activate.ps1
```

3. 패키지

```
python -m pip install --upgrade pip
python -m pip install -r .\Day01\배포용\requirements.txt
python -m pip check
python -c "import requests, pandas, jupyterlab; print('requests', requests.__version__); print('pandas', pandas.__version__); print('jupyterlab', jupyterlab.__version__)"
```

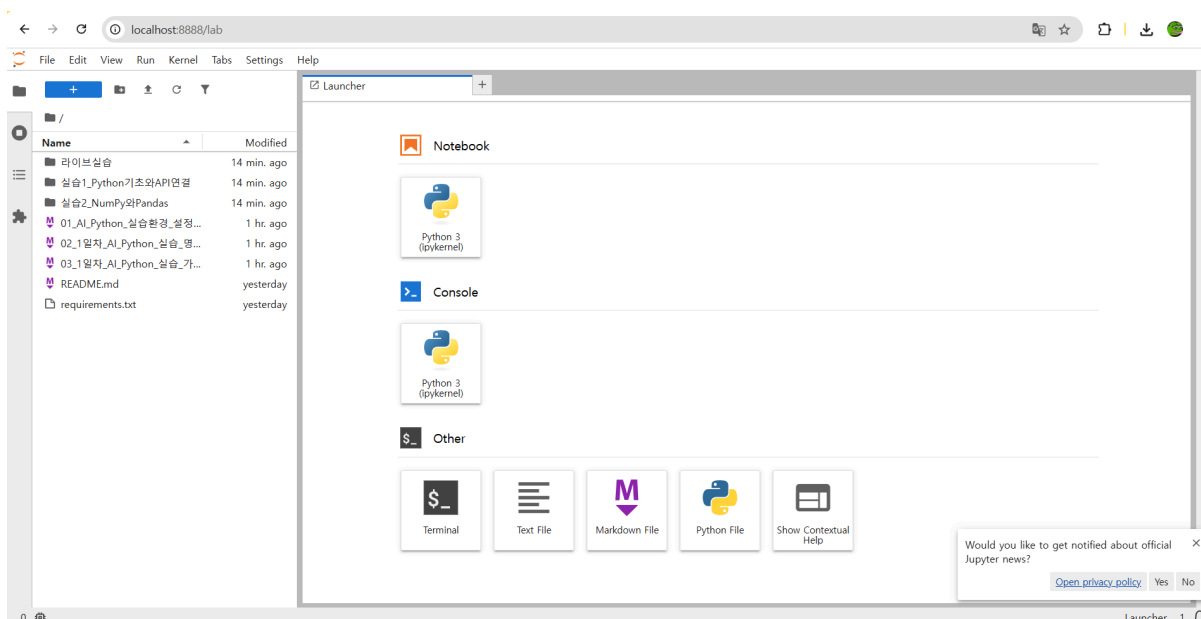
예상 출력: `No broken requirements found.`, requests, pandas, jupyterlab 버전

```
Windows PowerShell
, charset_normalizer, certifi, bleach, babel, attrs, async-lru, asttokens, terminado, stack_data, rfc3987-syntax, rfc3339-validator, requests, referencing, python-dateutil, prompt_toolkit, matplotlib-inline, jupyter-core, Jinja2, jedi, ipython-pygments-lexers, httpcore, contourpy, cffi, BeautifulSoup4, anyio, pandas, matplotlib, jupyter-server-terminals, jupyter-client, jupyter-builder, jsonschema-specifications, ipython, httpx, arrow, argon2-cffi-bindings, seaborn, jsonschema, isoduration, ipykernel, argon2-cffi, nbformat, nbclient, jupyter-events, nbconvert, jupyter-server, notebook-shim, jupyterlab-server, jupyter-lsp, jupyterlab
Successfully installed MarkupSafe-3.0.3 anyio-4.14.2 argon2-cffi-25.1.0 argon2-cffi-bindings-25.1.0 arrow-1.4.0 asttokens-3.0.2 async-lru-2.3.0 attrs-26.1.0 babel-2.18.0 BeautifulSoup4-4.15.0 bleach-6.4.0 certifi-2026.6.17 cffi-2.1.0 chardet-5.2.0 charset-normalizer-3.4.0 colorama-0.4.6 comm-0.2.3 contourpy-1.3.3 cyclopts-0.12.1 debugpy-1.8.21 decorator-5.3.1 defusedxml-0.7.1 executing-2.2.1 fastjsonschema-2.21.2 fonttools-4.63.0 fqdn-1.5.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.18 ipykernel-7.3.0 ipython-9.15.0 ipython-pygments-lexers-1.1.1 isoduration-20.11.0 jedi-0.20.0 Jinja2-3.1.6 json5-0.15.0 jsonpointer-3.1.1 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 jupyter-builder-1.1.1 jupyter-client-8.9.1 jupyter-core-5.9.1 jupyter-events-0.12.1 jupyter-lsp-2.3.1 jupyter-server-2.20.0 jupyter-server-terminals-0.5.4 jupyterlab-4.6.1 jupyterlab-pygments-0.3.0 jupyterlab-server-2.28.0 kiwisolver-1.5.0 lark-1.3.1 matplotlib-3.11.1 matplotlib-inline-0.2.2 mistune-3.3.3 nbclient-0.11.0 nbconvert-7.17.1 nbformat-5.10.4 nest-asyncio-1.7.2 notebook-shim-0.2.4 numpy-2.5.1 packaging-26.2 pandas-3.0.3 pandocfilters-1.5.1 parso-0.8.7 pillow-12.3.0 platformdirs-4.10.1 prometheus-client-0.25.0 prompt-toolkit-3.0.52 psutil-7.2.2 pure-eval-0.2.3 pycparser-3.0 pygments-2.20.0 pyparsing-3.3.2 python-dateutil-2.9.0.post0 python-jison-logger-4.1.0 pywinpty-3.0.5 pyyaml-6.0.3 pyzmq-27.1.0 referencing-0.37.0 requests-2.34.2 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.6.3 seaborn-0.13.2 send2trash-2.1.0 six-1.17.0 soupsieve-2.8.4 stack_data-0.6.3 terminado-0.18.1 tinycss2-1.5.1 tornado-6.5.7 traitlets-5.15.1 typing_extensions-4.16.0 tzdata-2026.3 uri-template-1.3.0 urllib3-2.7.0 wcwidth-0.8.2 webcolors-25.10.0 webencodings-0.5.1 websocket-client-1.9.0
(.venv) PS C:\ai-python> python -m pip check
No broken requirements found.
(.venv) PS C:\ai-python> python -c "import requests, pandas, jupyterlab; print('requests', requests.__version__); print('pandas', pandas.__version__); print('jupyterlab', jupyterlab.__version__)"
requests 2.34.2
pandas 3.0.3
jupyterlab 4.6.1
(.venv) PS C:\ai-python>
(.venv) PS C:\ai-python>
```

4. JupyterLab

`python -m jupyter lab .\Day01\배포용`

- 파일: `실습1_Python기초와API연결/실습1_Python기초와API연결_빈칸.ipynb`
- 커널: `Python 3`, `.venv`



5. API 연결

```
python -c "import requests; r=requests.get('https://jsonplaceholder.typicode.com/users', timeout=5); print(r.status_code, len(r.json()))"
```

- 정상 출력: 200 10
- 연결 실패: 샘플 사용자 데이터로 전환

```
(.venv) PS C:\ai-python> python -c "import requests; r=requests.get('https://jsonplaceholder.typicode.com/users', timeout=5); print(r.status_code, len(r.json()))"
200 10
```

6. API 키

- 기본 설정: API 키 불필요
- LLM 호출: mock_llm
- 확장 설정:

```
$env:OPENAI_API_KEY="발급받은_키"
```

- 노트북, 화면 캡처, Git 저장소에 키를 남기지 않습니다.
- .env 사용 시 .gitignore 에 .env 를 추가합니다.

오류 확인

증상	조치
python 을 찾을 수 없음	PATH 옵션을 선택해 Python을 설치하고 PowerShell을 새로 엽니다.
jupyter 를 찾을 수 없음	(.venv) 를 확인하고 requirements를 설치합니다.
ModuleNotFoundError	where.exe python 의 첫 경로를 확인합니다.
NameError	커널을 재시작하고 앞 셀부터 실행합니다.
AssertionError	assert 를 수정하지 않고 직전 코드의 None 을 채웁니다.

완료 확인

- ☐ Python 3.12.x와 설치 경로 확인
- ☐ ai-python/.venv 활성화
- ☐ requirements 설치와 pip check 완료

- ☐ 빈칸 노트북과 Python 3 커널 확인
- ☐ API 응답 또는 샘플 데이터 전환 확인
- ☐ 화면 캡처의 API 키 미노출 확인